Estimating mixing properties of local Hamiltonian dynamics and continuous quantum random walks is PSPACE-hard

Measuring local observables could probabilistically solve PSPACE

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Observable, Hamiltonian, spectral resolution

finite dimensional Hilbert space $\mathcal{H}$

observable / **Hamiltonian** : Hermitian operator A on $\mathcal{H}$

continuous quantum random walk : A is an adjacency matrix

spectral resolution of A:

- different eigenvalues of A: (λ)
- orthogonal projections onto the corresponding eigenspaces of A: (P_λ)

Mixing properties of Hamiltonian dynamics and continuous quantum random walks

Time-average of a state ρ on $\mathcal{H}$:

$$\bar{\rho} := \lim_{T\to\infty} \frac{1}{T} \int_{t=0}^{T} e^{-iAt}\, \rho\, e^{iAt}\, \mathrm{d}t$$

von-Neumann entropy $S(\bar{\rho})$ of the time-average

$$S(\bar{\rho}) := \mathrm{tr}(\bar{\rho} \log_2 \bar{\rho})$$

motivation for considering $S(\bar{\rho})$:

- quantifies “how much and how uniformly” the orbit $\rho_t := e^{-iAt}\, \rho\, e^{iAt}$ fills out the state space
- quantifies “how far” a pure state $\rho := |\Phi\rangle\langle\Psi|$ is from all eigenvectors

Measurement of observables

Postulate: Measurement of A with accuracy Δ:

for all states ρ on $\mathcal{H}$ the probability to obtain an outcome in the interval

$$[\lambda - \Delta, \lambda + \Delta]$$

is a least

$$(3/4)\,\mathrm{tr}(\rho P_\lambda)$$

Local operators

$\mathcal{H}$ tensor product space of n Hilbert spaces

$$\mathcal{H} := \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n$$

A is k-local:

A sum of operators acting on at most k tensor components non-trivially

example for 2-local Hamiltonians:

Ising interactions, coupling topology specified by a graph $G = (V, E)$

$$\sum_{(k,l)\in E} \sigma_z^{(k)} \sigma_z^{(l)}$$

$$\sigma_z^{(k)} = \mathbf{1} \otimes \cdots \otimes \mathbf{1} \otimes \underbrace{\sigma_z}_{k\text{th qubit}} \otimes \mathbf{1} \otimes \cdots \otimes \mathbf{1}$$

we want to prove

Estimating mixing properties of local Hamiltonian dynamics and continuous quantum random walks is PSPACE-hard

Measuring local observables could probabilistically solve PSPACE

Outline of the proof:

- recall definition of PSPACE (based on irreversible Turing machines)
- transform irreversible TMs to reversible TM
- construct quantum circuit that simulate the computational steps of the reversible TMs
- encode the sequence of gates of these quantum circuits in local operators

  $\longrightarrow$ desired observables, Hamiltonians, adjacency matrices

Definition of PSPACE

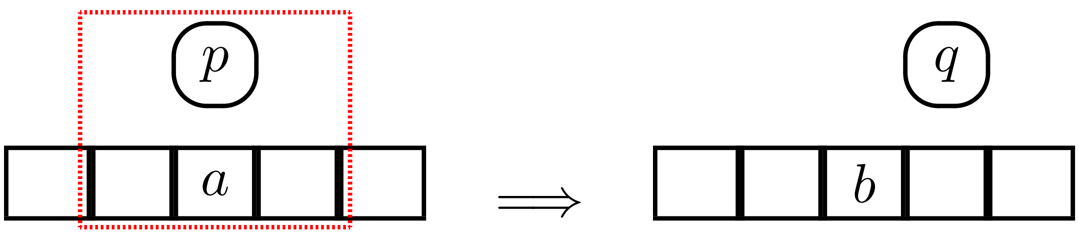
class of all languages recognizable by polynomial space bounded DTMs

Deterministic Turing Machine:

- head (internal states Q)

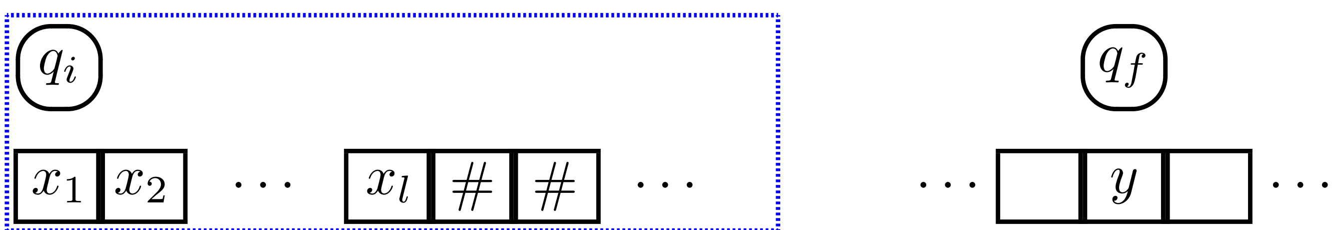
  tape divided into cells (contain symbols of the alphabet Σ)

- local transition function $Q \times \Sigma \rightarrow Q \times \Sigma \times \{-1, 0, +1\}$



- input: $x_1\, x_2 \ldots x_l$ output: symbol in current cell when RTM goes into a final state q_f

$poly(l)$



Quantified Boolean Formula Problem

is a PSPACE-complete problem

quantified Boolean formula:

$$Q_1 v_1\, Q_2 v_2 \cdots Q_n v_n\, B(v_1, v_2, \ldots, v_n)$$

Q_i: quantifiers (either $\forall$ or $\exists$)

$B(v_1, v_2, \ldots, v_n)$: Boolean formula (without quantifiers) in the variables $v_1, v_2, \ldots, v_n$

quantified Boolean formula problem:

determine if a quantified formula is true

it can be solved within space $O(l^2)$ (l input length) by an irreversible TM

Reversible Turing machines

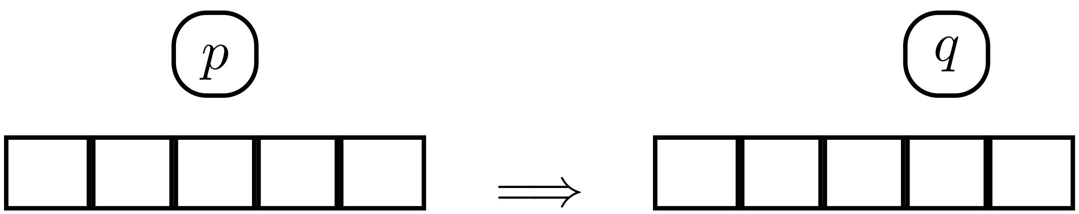
Lange et al., "Reversible space equals deterministic space", J. Comp. & Syst. Sci., 2000.

irreversible TMs can be simulated space-efficiently by reversible TMs

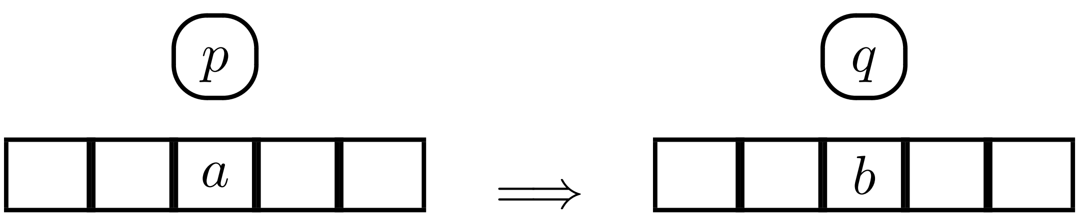
explicit construction of the RTM

each transition is

- either moving $p \to (q, \pm 1)$



- or read-and-write $(p, a) \to (q, b)$



Quantum circuit simulating the RTM (first idea)

registers: `head`, `tape_index`, `ACC`, `cell[i]` $i = 0, \ldots, N$

- U_{moving} implements $p \to (q, \pm 1)$

$$|p\rangle_{\text{head}} \otimes |i\rangle_{\text{tape\_index}} \to |q\rangle_{\text{head}} \otimes |i \pm 1\rangle_{\text{tape\_index}}$$

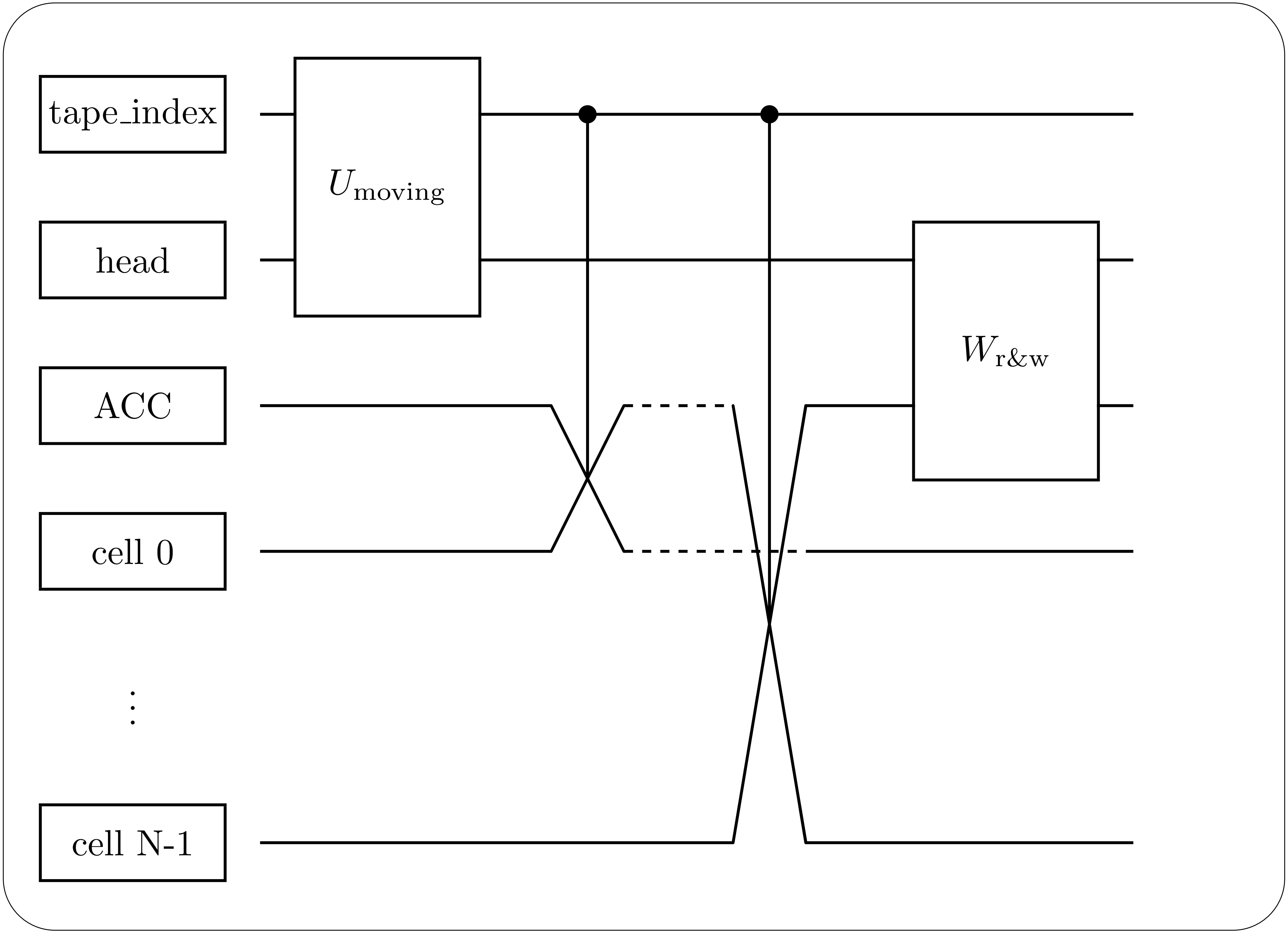
- $U_{\text{r\&w}}$ implements $(p, a) \to (q, b)$
  - swap `ACC` and `cell[i]` (current cell)

$$\prod_{i=0}^{N-1} \Lambda_i \big(\text{swap}(\text{ACC}, \text{cell[i]})\big) \qquad \Lambda_i \text{ applies swap iff } \texttt{tape\_index}{=}\text{i}$$

  - apply $W_{\text{r\&w}}$

$$|p\rangle_{\text{head}} \otimes |a\rangle_{\text{ACC}} \to |q\rangle_{\text{head}} \otimes |b\rangle_{\text{ACC}}$$

  - swap `ACC` and `cell[i]`



tape_index
head
ACC
cell 0
⋮
cell N-1
U_{moving}
$W_{\mathrm{r\&w}}$

Disadvantages of the quantum circuit

- not known exactly: how many times U should be applied

  desired: running time r_l depends only on input length l

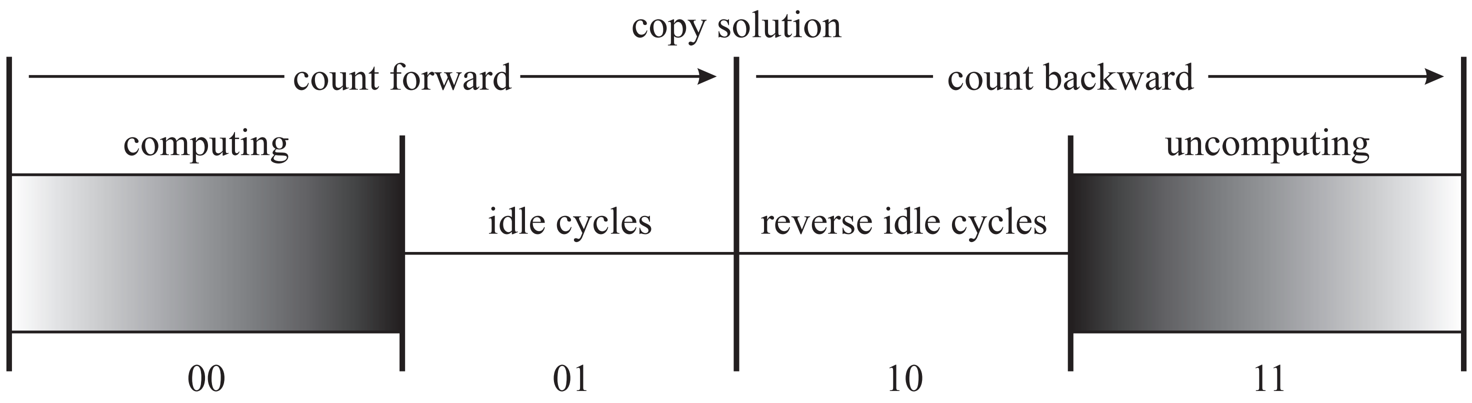
- U produces garbage (e.g. does not reset ancillas)

  desired:

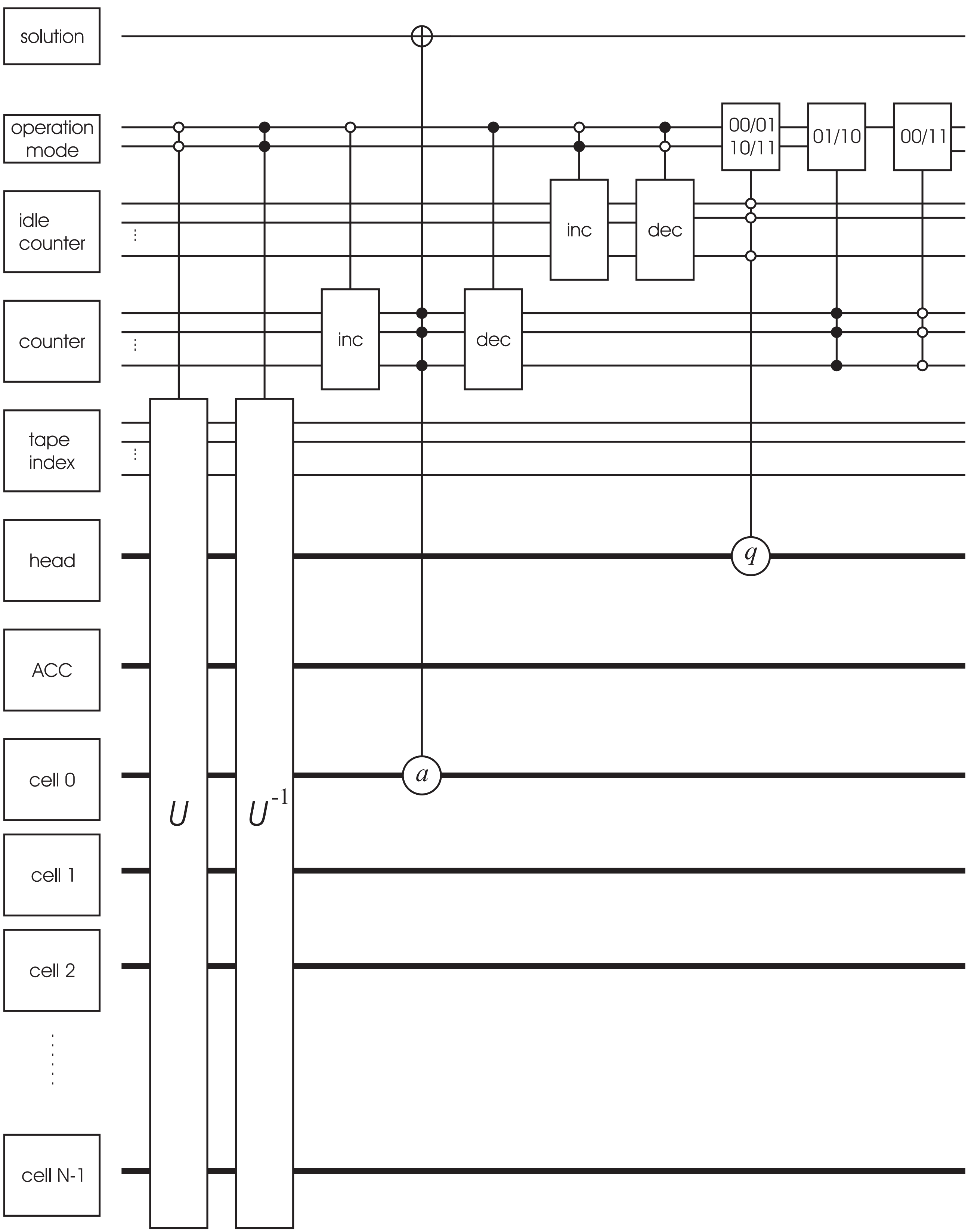
$$V^{r_l}|x\rangle \otimes |y\rangle \otimes |00\ldots 0\rangle = |x\rangle \otimes |f(x) \oplus y\rangle \otimes |00\ldots 0\rangle$$

$f(x)$ solution to the instance x

$\Longrightarrow$ second idea: augment U and $U^\dagger$ with some control logic



Final quantum circuit



Characterization of PSPACE with quantum circuits

L arbitrary language in PSPACE

$$\Downarrow$$

family $(V_l)_{l\in\mathbb{N}}$ of quantum circuits such that

V_l decides for all input x of length l if $x \in L$:

$$V_l^{r_l}(|x\rangle \otimes |y\rangle \otimes |00\ldots0\rangle) = |x\rangle \otimes |y \oplus f(x) \bmod 2^{c_l}\rangle \otimes |00\ldots0\rangle\,,$$

properties of V_l:

V_l permutes only computational basis states

powers of V_l produce orthogonal states (until recurrence)

$f(x) = 0 \qquad |\Psi_x\rangle\,, V_l|\Psi_x\rangle\,, \ldots, V_l^{r_l-1}|\Psi_x\rangle\,, V_l^{r_l}|\Psi_x\rangle = |\Psi_x\rangle$

$f(x) = 1 \qquad |\Psi_x\rangle\,, V_l|\Psi_x\rangle\,, \ldots, V_l^{2^{c_l} r_l-1}|\Psi_x\rangle\,, V_l^{2^{c_l} r_l}|\Psi_x\rangle = |\Psi_x\rangle$

$\Longrightarrow$ orbit length depends on $x \in L$

Local Hamiltonians/observables encoding the circuits

T_k: elementary quantum gates (Toffoli gates) of V $(k = 0, \ldots, s-1)$

`clock`: quantum register on s qubits

$$\rightarrow |1000\ldots0\rangle \mapsto |0100\ldots0\rangle \mapsto |0010\ldots0\rangle \mapsto \ldots \mapsto |000\ldots01\rangle \vdash$$

forward-time operator

$$\begin{aligned} F &= T_0 \quad \otimes |1\rangle_2\langle 0|_1 \otimes |0\rangle_1\langle 1|_0 + T_1 \otimes |1\rangle_3\langle 0|_2 \otimes |0\rangle_2\langle 1|_1 + \cdots + \\ & \quad T_{s-1} \otimes |1\rangle_1\langle 0|_0 \otimes |0\rangle_s\langle 1|_{s-1} \end{aligned}$$

local Hamiltonian/observable

$$A := F + F^\dagger$$

Orbit properties

- F acts as a cyclic shift on $\mathcal{O} := \left\langle F^j |\Psi_x\rangle \mid j \in \mathbb{N} \right\rangle$
- dimension of $\mathcal{O}$ is $d := 2^{c\, f(x)}\, r\, s$ depends on the solution

$\Longrightarrow$ eigenvalues and eigenvectors of F restricted to $\mathcal{O}$ are

$$e^{2\pi k/d} \qquad |k\rangle \qquad \text{for } k = 0, \ldots d-1$$

$\Longrightarrow$ $|\Psi_x\rangle$ uniform superposition of all eigenvectors $|k\rangle$

$$|\Psi_x\rangle := \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} |k\rangle$$

- F and $F^\dagger$ commute on $\mathcal{O}$

$\Longrightarrow$ eigenvalues and eigenvectors of A restricted to $\mathcal{O}$ are

$$2\cos(2\pi k/d) \qquad |k\rangle \qquad \text{for } k = 0, \ldots d-1$$

2 EVs with multiplicity 1 ($\equiv$ real EVs of F)

$d-2$ EVs with multiplicity 2 ($\equiv$ complex EVs of F)

Entropy of time-average

initial state:

$$|\Psi_x\rangle\langle\Psi_x| = \frac{1}{d}\sum_{k,l} |k\rangle\langle l|$$

averaging over time destroys coherence between eigenspaces

$$\bar{\rho} := \frac{1}{d}\sum_{k,l} [\cos(2\pi k/d) = \cos(2\pi l/d)]\, |k\rangle\langle l| = \begin{pmatrix} 1 & & & & & & \\ & 1 & & & & & \\ & & 1 & 1 & & & \\ & & 1 & 1 & & & \\ & & & & \ddots & & \\ & & & & & 1 & 1 \\ & & & & & 1 & 1 \end{pmatrix}$$

entropy $S(\bar{\rho}) = \log_2 d - (d-2)/d$

d depends on whether x is a "yes" or "no" instance

$\Longrightarrow$ difference in entropy between "yes" and "no" instance at least $c/2$ bits

c size of the output register

Measurement of A

output register consists of one qubit

prepare $|\Psi_x\rangle$

perfect measurement of A yields the results

- $2\cos(2\pi k/(2rs)), \quad k = 0, \ldots 2rs - 1$, if x "yes" instance
- $2\cos(2\pi k/(\ rs)), \quad k = 0, \ldots\ \ rs - 1$, if x "no" instance

with equal probability

$\Longrightarrow$ we can distinguish between both cases after few samples

imperfect measurement

accuracy in the order $1/(rs)$ is sufficient to distinguish between both cases